

Interactive divisibility graph number reconstruction simulator

Application example

NUMBERS THAT GENERATE THE WALK ' $0 \rightarrow 2 \rightarrow 1 \rightarrow 3$ '
USING THE DIVISIBILITY GRAPH MODULO 7

This document shows, through a practical example, how to use the interactive number reconstruction simulator based on divisibility graphs. For this purpose, the sequence of states $0 \rightarrow 2 \rightarrow 1 \rightarrow 3$ and the divisibility graph modulo 7 are used as an example.

This resource complements the «Interactive graph-based activity for divisibility». Unlike the conventional simulator, in which a complete number is entered and the corresponding walk is observed, this tool approaches the process in the reverse direction: the user defines a modulus and constructs a walk between states of the graph. Based on this walk, the simulator determines which digits can produce each transition and, when the walk is complete, reconstructs all numbers compatible with the indicated sequence of states.

This resource makes it possible to explore divisibility not only as a calculation procedure, but also as a problem of reconstruction, reasoning, analysis of possibilities, and the search for solutions within a graph.



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USER MANUAL

STEP 1. SELECTION OF THE MODULUS AND THE WALK LENGTH

The first step is to select the modulus to be used. The simulator supports moduli ranging from 2 to 20. Next, the length of the walk to be constructed must be specified. The length represents the number of transitions in the walk and can take values between 1 and 50. For example, for a walk length of 3, a sequence consisting of four states will be generated, since there are three transitions between four consecutive states.

Once both values have been entered, the «Create walk» button must be pressed. The simulator will automatically generate the necessary fields for entering the walk and construct the graph corresponding to the selected modulus (see Figure 1).

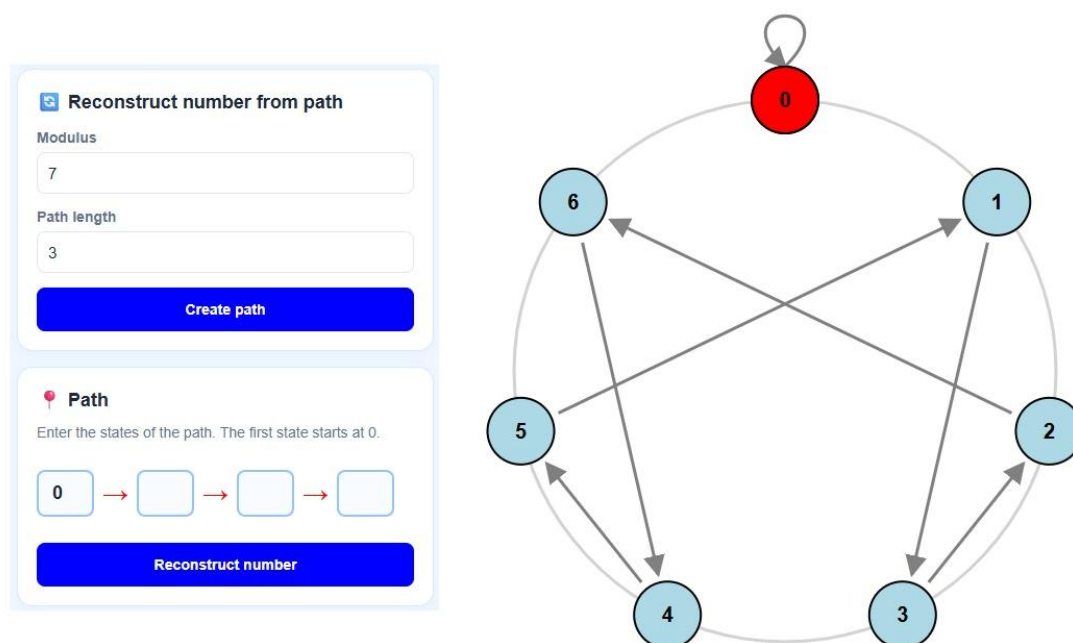


Figure 1. Simulator interface after selecting the modulus and the length of the walk.

STEP 2. WALK CONSTRUCTION AND TRANSITION ANALYSIS

The first state of the walk is automatically set to 0, since the processing of a number begins with a remainder equal to zero before its first digit is incorporated (see Figure 1). The remaining states must be entered manually by the user.

Each state entered must be between 0 and ' $m-1$ ', where ' m ' is the selected modulus. The simulator validates the entered values and visually indicates those that do not belong to the permitted range.

One of the main functions of the simulator is to determine which digits can produce each transition in the walk. To do this, the system analyzes each pair of consecutive states. If a transition is $r_i \rightarrow r_{i+1}$, the simulator searches for all digits between 0 and 9 that satisfy the condition $(10 \cdot r_i + d) \bmod n = r_{i+1}$. The digits that satisfy this equality are the possible digits that can produce the transition.

This information is displayed in the interface for each step of the walk. For example, if a transition $2 \rightarrow 1$ admits the digits 2 and 9, the simulator will indicate that these two digits are compatible with that transition. This makes it possible to observe how the same sequence of states can correspond to different sequences of digits.

As the states are entered, the walk is automatically represented on the graph using red arrows. In this way, the user can simultaneously observe the sequence of states being constructed and its graphical representation. Figure 2 shows the steps of the sequence of states $0 \rightarrow 2 \rightarrow 1 \rightarrow 3$ using the divisibility graph modulo 7.

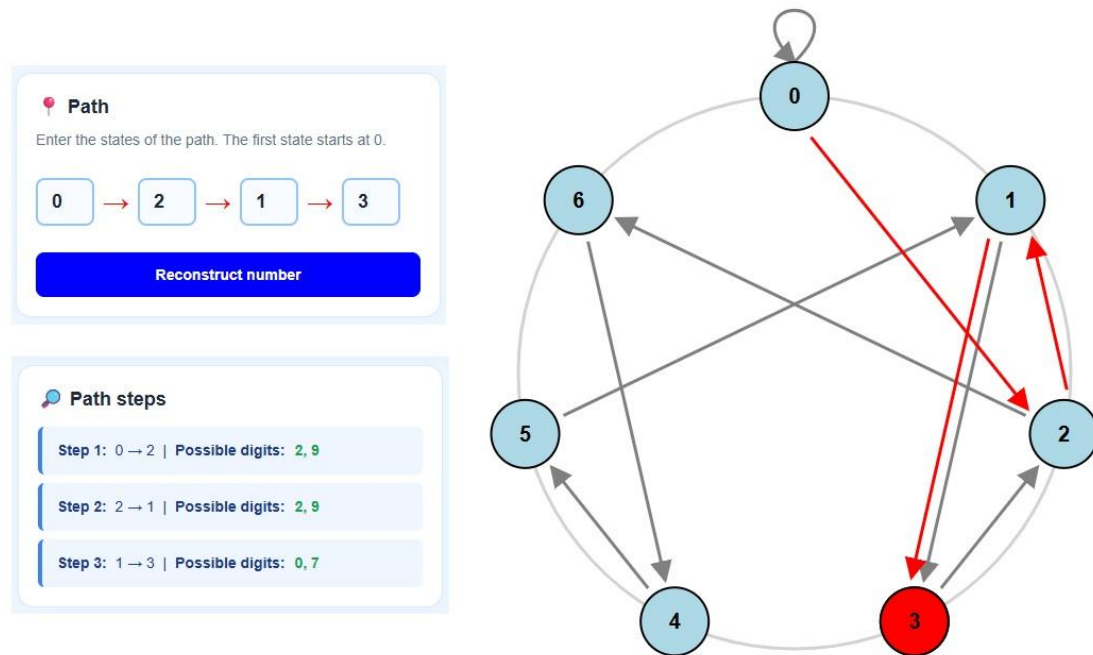


Figure 2. Representation of the walk $0 \rightarrow 2 \rightarrow 1 \rightarrow 3$ on the divisibility graph modulo 7, including the possible digits for each transition.

Not all sequences of states are possible for a given modulus. When the simulator analyzes a transition, it checks whether there is at least one digit between 0 and 9 that allows the transition to occur. If there is no compatible digit, the system informs the user that the transition is not valid for the selected modulus. In this case, the walk cannot be used to reconstruct a number, and one of the states involved must be modified.

For example, for modulus 13, the transition $2 \rightarrow 6$ is not possible, since there is no digit 'd' between 0 and 9 that satisfies the condition $(10 \cdot 2 + d) \bmod 13 = 6$. The application will display the message "No possible digit" (see Figure 3). The user can use the vertical control

located in the walk fields to modify their value. The application will automatically update the possible digits corresponding to each transition. This functionality also allows the simulator to be used as an exploration tool.



Path

Enter the states of the path. The first state starts at 0.

0 → 2 → 6 → []

Reconstruct number

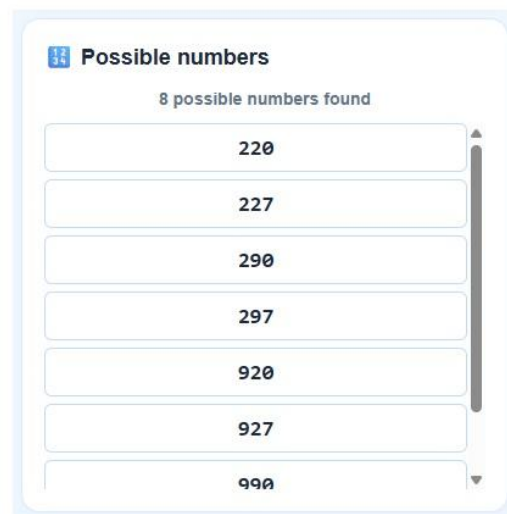
Path steps

Step 1: 0 → 2 | Possible digits: 2

Step 2: 2 → 6 | No possible digit

Figure 3. Detection of an invalid transition in the divisibility graph modulo 13: no digit between 0 and 9 can produce the transition $2 \rightarrow 6$.

When all the walk fields have been completed, the user can click the «Reconstruct number» button. The simulator then uses the possible digits for each transition to generate all combinations compatible with the complete walk. In the example in Figure 2, each of the three transitions in the walk admits 2 possible digits. Consequently, the system will generate the $2 \cdot 2 \cdot 2 = 8$ possible combinations. Each combination represents a number whose digits produce exactly the indicated sequence of states. The result is displayed in the interface together with the total number of solutions found.



Possible numbers

8 possible numbers found

220

227

290

297

920

927

990

Figure 4. Possible numbers reconstructed from the walk $0 \rightarrow 2 \rightarrow 1 \rightarrow 3$ in the divisibility graph modulo 7.

During the reconstruction process, combinations whose first digit is 0 may appear. Although these combinations may correctly satisfy the mathematical conditions of the walk, they are not considered numbers of the corresponding length when they begin with zero. For this reason, the simulator automatically removes these combinations from the final results. It also informs the user of the number of solutions that have been discarded because they begin with zero.

For example, if, for modulus 7, the walk were $0 \rightarrow 0 \rightarrow 1 \rightarrow 3$, the possible digits generated by the first transition would be 0 and 7. In this case, the system would remove the combinations beginning with 0 (see Figure 5).



Figure 5. Removal of reconstructed combinations beginning with zero from the final set of possible numbers.

STEP 3. COMPLETION AND NEW EXERCISE

Once the reconstruction has been completed, the user can analyze the results obtained or use the simulator again to study another walk. If another walk using the same modulus and the same length is desired, the «Create walk» option can be used again. When starting a completely new activity, the «New exercise» button should be used. This option resets the modulus and length fields, removes the previous walk, clears the reconstruction results, and clears the graphical representation.

Usage recommendations

It is recommended to begin with small moduli and transitions in order to become familiar with the operation of the divisibility graph and then progressively increase the difficulty. It is also advisable to first analyze the possibilities for each transition before carrying out the complete reconstruction, as this information helps explain why certain numbers appear among the solutions. When a transition is not valid, it is advisable to first identify the specific step causing the problem and mathematically verify whether there is any digit that satisfies the modular relationship. This makes it possible to use the simulator as a reasoning tool rather than merely as an automatic answer generator.